



SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 25 / 25 completed

Remove

REMOVE

Write a C program to remove an element from an array.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, pos;
5
6     // Input number of elements
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    int arr[100];
11
12    // Input array elements
13    printf("Enter array elements:\n");
14    for (int i = 0; i < n; i++) {
15        scanf("%d", &arr[i]);
16    }
17
18    // Input position to delete
19    printf("Enter position to delete (1 to %d): ", n);
20    scanf("%d", &pos);
21}
```

OUTPUT

```
Enter number of elements: Enter array elements:
Enter position to delete (1 to 3): Array after deletion:
1 3
```

INPUT

```
3
1 2 3
2
```

QUESTIONS 25 / 25 completed

Palindrome

PALINDROME

Write a C program to check if an array is a palindrome or not.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, isPalindrome = 1;
5
6     // Input size of array
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    int arr[n];
11
12    // Input array elements
13    printf("Enter array elements:\n");
14    for (int i = 0; i < n; i++) {
15        scanf("%d", &arr[i]);
16    }
17
18    // Check palindrome
19    for (int i = 0; i < n / 2; i++) {
20        if (arr[i] != arr[n - i - 1]) {
21            isPalindrome = 0;
22        }
23    }
24}
```

OUTPUT

```
Enter number of elements: Enter array elements:
Array is a palindrome.
```

INPUT

```
5
1 2 1 2 1
```

QUESTIONS 25 / 25 completed

Add 2 Matrix

ADD 2 MATRIX

Write a C program to add two matrices.

PROGRAM EDITOR

Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[100][100],b[100][100];
5     int i,j,m,n;
6     int sum[100][100];
7
8     printf("enter number of rows and columns:");
9     scanf("%d %d",&m,&n);
10
11     printf("first matrix:\n");
12     for(i=0;i<m;i++)
13     {
14         for(j=0;j<n;j++)
15         {
16             scanf("%d",&a[i][j]);
17         }
18     }
19 }
20
21 printf("second array:\n");
```

OUTPUT

```
enter number of rows and columns: first matrix:
second array:
addition =
7 7
5 7
```

INPUT

```
2 2
1 2
4 5
6 5
1 2
```

QUESTIONS 25 / 25 completed

Transpose

TRANSPOSE

Write a C program to find the transpose of a matrix.

PROGRAM EDITOR

Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[100][100];
5     int i,j,m,n;
6     int trans[100][100];
7
8     printf("enter number of rows and columns:");
9     scanf("%d %d",&m,&n);
10
11     printf("first matrix:\n");
12     for(i=0;i<m;i++)
13     {
14         for(j=0;j<n;j++)
15         {
16             scanf("%d",&a[i][j]);
17         }
18     }
19 }
20
21
```

OUTPUT

```
enter number of rows and columns: first matrix:
transpose =
1 3
2 4
```

INPUT

```
2 2
1 2
3 4
```



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QUESTIONS 25 / 25 completed

Product Diag. ▾

PRODUCT DIAG.

Write a C program to find the product of the diagonal elements of a matrix.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int rows, cols;
5     int product = 1;
6
7     // Input rows and columns
8     printf("Enter number of rows and columns: ");
9     scanf("%d %d", &rows, &cols);
10
11     int matrix[rows][cols];
12
13     // Input matrix elements
14     printf("Enter matrix elements:\n");
15     for (int i = 0; i < rows; i++) {
16         for (int j = 0; j < cols; j++) {
17             scanf("%d", &matrix[i][j]);
18         }
19     }
20
21     // Find product of diagonal elements
```

OUTPUT

Running...

INPUT

2 2
1 2
3 4

QUESTIONS 25 / 25 completed

Sum Low Tri. ▾

SUM LOW TRI.

Write a C program to find the sum of the lower triangular elements of a matrix.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, sum = 0;
5
6     // Input size of square matrix
7     printf("Enter order of matrix: ");
8     scanf("%d", &n);
9
10    int matrix[n][n];
11
12    // Input matrix elements
13    printf("Enter matrix elements:\n");
14    for (int i = 0; i < n; i++) {
15        for (int j = 0; j < n; j++) {
16            scanf("%d", &matrix[i][j]);
17        }
18    }
19
20    // Find sum of lower triangular elements
21    for (int i = 0; i < n; i++) {
```

OUTPUT

```
Enter order of matrix: Enter matrix elements:
Sum of lower triangular elements = 75
```

INPUT

```
3 3
1 2 3
4 5 6
```

QUESTIONS 25 / 25 completed

Sum Right Diag. ▾

SUM RIGHT DIAG.

Write a C program to find the sum of the elements in the right diagonal of a matrix.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, sum = 0;
5
6     // Input order of matrix
7     printf("Enter order of matrix: ");
8     scanf("%d", &n);
9
10    int matrix[n][n];
11
12    // Input matrix elements
13    printf("Enter matrix elements:\n");
14    for (int i = 0; i < n; i++) {
15        for (int j = 0; j < n; j++) {
16            scanf("%d", &matrix[i][j]);
17        }
18    }
19
20    // Find sum of right diagonal elements
21    for (int i = 0; i < n; i++) {
```

OUTPUT

Running...

INPUT

3 3
1 2 3
4 5 6



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QUESTIONS 25 / 25 completed

Diagonal or Not

DIAGONAL OR NOT

Write a C program to check if a matrix is diagonal or not.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, isDiagonal = 1;
5
6     // Input order of square matrix
7     printf("Enter order of matrix: ");
8     scanf("%d", &n);
9
10    int matrix[n][n];
11
12    // Input matrix elements
13    printf("Enter matrix elements:\n");
14    for (int i = 0; i < n; i++) {
15        for (int j = 0; j < n; j++) {
16            scanf("%d", &matrix[i][j]);
17        }
18    }
19
20    // Check diagonal matrix
21    for (int i = 0; i < n; i++) {
```

OUTPUT

```
Enter order of matrix: Enter matrix elements:
Matrix is not a diagonal matrix.
```

INPUT

```
2 2
1 2
4 5
```


QUESTIONS 25 / 25 completed

Low Tri. Not

LOW TRI. NOT

Write a C program to check if a matrix is lower triangular or not.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, isLower = 1;
5
6     // Input order of square matrix
7     printf("Enter order of matrix: ");
8     scanf("%d", &n);
9
10    int matrix[n][n];
11
12    // Input matrix elements
13    printf("Enter matrix elements:\n");
14    for (int i = 0; i < n; i++) {
15        for (int j = 0; j < n; j++) {
16            scanf("%d", &matrix[i][j]);
17        }
18    }
19
20    // Check lower triangular matrix
21    for (int i = 0; i < n; i++) {
```

OUTPUT

```
Enter order of matrix: Enter matrix elements:
Matrix is not a lower triangular matrix.
```

INPUT

```
22
15
49
```

QUESTIONS 25 / 25 completed

Equal

EQUAL

Write a C program to check if two matrices are equal or not.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int rows, cols, equal = 1;
5
6     // Input rows and columns
7     printf("Enter number of rows and columns: ");
8     scanf("%d %d", &rows, &cols);
9
10    int a[rows][cols], b[rows][cols];
11
12    // Input first matrix
13    printf("Enter elements of first matrix:\n");
14    for (int i = 0; i < rows; i++) {
15        for (int j = 0; j < cols; j++) {
16            scanf("%d", &a[i][j]);
17        }
18    }
19
20    // Input second matrix
21    printf("Enter elements of second matrix:\n");
```

OUTPUT

```
Enter number of rows and columns: Enter elements of first matrix:
Enter elements of second matrix:
Matrices are not equal.
```

INPUT

```
2 2
4 6
9 6
2 2
4 6
9 6
```



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QUESTIONS 25 / 25 completed

Multiply Scalar

MULTIPLY SCALAR

Write a C program to multiply a matrix by a scalar.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int rows, cols, scalar;
5
6     // Input rows and columns
7     printf("Enter number of rows and columns: ");
8     scanf("%d %d", &rows, &cols);
9
10    int matrix[rows][cols];
11
12    // Input matrix elements
13    printf("Enter matrix elements:\n");
14    for (int i = 0; i < rows; i++) {
15        for (int j = 0; j < cols; j++) {
16            scanf("%d", &matrix[i][j]);
17        }
18    }
19
20    // Input scalar value
21    printf("Enter scalar value: ");
```

OUTPUT

Running...

INPUT

2 2
1 2
4 5
4

QUESTIONS 25 / 25 completed

Inverse

INVERSE

Write a C program to find the inverse of a 2x2 matrix.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     float matrix[2][2], inverse[2][2];
5     float determinant;
6
7     // Input matrix elements
8     printf("Enter elements of 2x2 matrix:\n");
9
10    for (int i = 0; i < 2; i++) {
11        for (int j = 0; j < 2; j++) {
12            scanf("%f", &matrix[i][j]);
13        }
14    }
15
16    // Calculate determinant
17    determinant = (matrix[0][0] * matrix[1][1]) -
18                  (matrix[0][1] * matrix[1][0]);
19
20    // Check if inverse exists
21    if (determinant == 0) {
```

OUTPUT

```
Enter elements of 2x2 matrix:
Inverse matrix:
1.00 -1.00
-0.50 1.00
```

INPUT

```
2 2
1 2
4 5
```



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Gauss-Jordan

Write a C program to perform Gauss-Jordan elimination on a matrix.

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5
6     // Input order of matrix
7     printf("Enter order of matrix: ");
8     scanf("%d", &n);
9
10    float a[n][2 * n];
11
12    // Input augmented matrix [A|I]
13    printf("Enter matrix elements:\n");
14
15    for (int i = 0; i < n; i++) {
16        for (int j = 0; j < n; j++) {
17            scanf("%f", &a[i][j]);
18        }
19
20        // Initialize identity matrix
21        for (int j = n; j < 2 * n; j++) {
```

```
Enter order of matrix: Enter matrix elements:
Matrix after Gauss-Jordan elimination:
1.00 0.00 -2.00 1.00
-0.00 1.00 1.25 -0.50
```

22
45
89